Scientometrics (2016) 109:1097–1116
DOI 10.1007/s11192-016-2074-3

Relation between awarding of Grants-in-aid for scientific research and characteristics of applicants in Japanese universities

Moritaka Hosotsubo[1,2] · **Ryuei Nishii**[2]

Received: 16 March 2016 / Published online: 30 July 2016

Abstract Japan's system of Grants-in-aid for scientific research aims to promote creative and pioneering research across a wide spectrum of fields, ranging from the humanities and social sciences to the natural sciences. The grants, amounting to 191 billion JPY (in 2006 FY) per year, are the main sources of competitive research funds for applicants belonging to national and public universities. In this paper, we examine this system by performing a statistical analysis on the relation between the awarding of grants and the attributes of applicants, including field, degree, position, and other covariates. We used Poisson, negative binomial distributions, their generalized linear models and logistic regressions for implementing covariates to the model. The analysis reveals interesting features of academic fields as well as the attributes of the awarded applicants.

Keywords Difference of academic fields · Model selection · Multinomial logistic regression · Zero-inflated distribution

Mathematics Subject Classification 62

JEL Classification C81

Introduction

Japan's system of Grants-in-aid for scientific research called KAKEN (the grants, hereafter) aims to promote creative and pioneering research across a wide spectrum of fields, ranging from the humanities and social sciences to the natural sciences. The grants,

✉ Moritaka Hosotsubo
hosotubo@nistep.go.jp

1 National Institute of Science and Technology Policy, MEXT, 3-2-2 Kasumigaseki, Chiyoda-ku, Tokyo 100-0013, Japan

2 Institute of Mathematics for Industry, Kyushu University, 744 Motooka, Nishi-ku, Fukuoka 819-0395, Japan

amounting to 191 billion JPY (in 2006 FY) per year, are the main sources of competitive research funds for applicants belonging to national and public universities. In this paper, we examine the relation between awarding of the grants and personal characteristics (attributes) of applicants via statistical methods, e.g., Poisson, negative binomial distributions, their generalized linear models and logistic regression.

Statistical studies on scientific research productivity by Rao (1980) and Yamasaki (1982) claimed that a negative binomial distribution provided the best fit for the relation between the number of papers and number of scholars. The grants in Japan have been studied and analyzed at the Japan Society for the Promotion of Science (JSPS), National Institute of Informatics (NII), and other institutions. For example, Ida and Fukuzawa (2013) examined the effects of funding on the publication of papers in Japan. Fukuzawa (2014) also studied the characteristics of investigators and research productivity. Here, we assume that there exists a relation between awarding and characteristics of applicants. We note, however, that awarding of the grants does not in itself relate to the productivity of scientific research.

Databases of the grants are available from KAKEN (https://kaken.nii.ac.jp/en/index). They describe research projects in terms of fields, institutions, names of investigators, and periods of projects. There are, however, no further personal characteristics of the investigators. The most important point is that the KAKEN database is only for chosen projects. Therefore, an analysis including faculty members who did not apply or failed to get the grants is impossible on the database. This is the reason why there is little research on the relation between the awarding of grants and the researchers' characteristics.

In this paper, we derive a statistical model illustrating the relation between awarding the grant and the researchers' personal characteristics such as graduating universities, degrees and number of transfers, which are not available from the KAKEN database. Our approach is new in the sense that the personal characteristics of researchers who failed to get the grants are included as the covariates of the statistical model. Furthermore, a follow-up study was conducted from 1988 to 2006, in which transfers of faculty members were traced by using the Higher-education Teacher's Mobility Data Base (HM-DB).

HM-DB recorded the transfers and promotions of faculty members at Japanese national universities. It compiled information from nine points in time (1988, 1991, 1994, 1997, 2000, 2003, 2005, 2006, 2008) from published information such as the Japanese University Employee List (Hosotsubo 2011). The quality of the data examined by the author as "The data, while having a somewhat low rate of determining the position of assistant professors, have a high rate of determining other higher positions". This means the quality of the data is sufficient for making comparisons with public statistics.

This paper is organized as follows. Section "Attributes of applicants and abbreviations" describes the explanatory variables (covariates) of faculty members. Section "Distributions of awarded grants" estimates the zero-inflated distributions of the reward numbers in all fields. On the basis of the covariates of faculty members, the number of rewards in all fields is modeled using zero inflated distributions and multinomial logistic distributions. Then, the optimal model is selected using the Bayesian information criterion (BIC). In addition, the levels of each covariate are aggregated if the aggregation improves the BIC value. Section "Optimal model by field" gives a discussion by field and clarifies the features of the respective academic fields. Section "Concluding remarks" concludes the paper. Appendices 1 and 2 give additional and detailed analyses on all fields and the individual fields, respectively.

Attributes of applicants and abbreviations

The transfers and promotions of faculty members at Japanese national universities were statistically analyzed using HM-DB (Hosotsubo 2011).

In this study, we connected HM-DB data with the number of scientific research grants awarded. We should point out here that the Japanese system of the grants has evolved over the years. Here, we treated the number of time points in which a faculty member was selected as a representative researcher of the grants and the target variable to be predicted. Furthermore, we created a total of 13 categories of covariates from 110,056 samples. See Table 1 for the covariates and levels. Note that universities G1, G2, G3, and G4 of [Affiliation] in Table 1 are an ordered group of national universities based on research productivity (G1, consisting of four universities, is the most active group, G2, consisting of ten universities, comes next, and so on); see Hosotsubo (2011) for details.

For example, the characteristics (covariates) of a certain researcher belonging to a national university in 1988–2006 are given as Fy: 2006, #op: 8, Af: G2, Riu: Kyushu, Ps: Prof., Gn: T4, Dg: Doctor, Gu: G2, Rigu: Kyushu, Fl: Mathematics, #t: 1 and #p: 1. If he/she was awarded four times in the eight observed time points, 4 is the value of the target variable.

Distributions of awarded grants

Figure 1 shows the frequencies of the number of awarded grants for the eight time points by academic field. A monotone decreasing tendency is apparent for each field. First, we fitted Poisson (PO) and negative Binomial (NB) distributions to the counting data. Unfortunately, the fit is not satisfactory because there are an excess of zero-count data. Therefore, we fitted zero-inflated Poisson (ZIP) and zero-inflated NB (ZINB) distributions to evaluate the zero-excess probabilities. See "Comparison of distributions of number of awards in all fields and in each field" for details on the zero-inflated distribution. We utilized R Packages gamlss (Stasinopoulos et al. 2015) and nnet (Ripley and Venables 2015) for numerical computation.

Features of eleven academic fields in terms of zero-inflation probability

We selected the best models by field by using the Akaike information criterion (AIC) and tabulate them in Table 2. It can be seen that the zero-inflated distributions are selected in all of the fields. For some fields, ZIP with two degrees of freedom (df, number of parameters) is superior to ZINB with 3 df. In addition, the zero-inflation probability is heavily dependent on the field of specialization. The three largest zero-inflation probabilities are denoted in bold face. In particular, Humanities and Social Sciences have large zero-inflate probabilities. The difference in these cases comes from the fact that these fields are not experimental or theoretical sciences. Note that the HM-DB analysis did not show such a difference.

The zero-inflation probability of Medicine/Dentistry/Pharmacology is fairly large because the field includes physicians of university-affiliated hospitals. Japanese university hospitals belong to universities and some of their physicians only perform treatments of patients; i.e., they do not also do research. Another reason is that the Japanese Ministry of Health, Labour, and Welfare supports a similar grant system called the Health Labour Sciences Research Grant. Thus, a number of researchers in this field are awarded these grants.

Table 1 Categories of professors' covariates, their abbreviations, and levels

Abbrevi.	Category of explanatory variables	Levels	# of levels
Fy	[Final year] the final year of the observation	1988, 1991, 1994, 1997, 2000, 2003, 2005, 2006	8
#op	[# of observation points] number of time points until end of observation	1, 2, 3, 4, 5, 6, 7, 8	8
Af	[Affiliation]	G1, G2, G3, G4, Public univ., Private univ., National academy/other	7
Riu	[Region in univ.]	Kanto, Chubu, Kinki, Chugoku/Shikoku, Hokkaido/Tohoku, Kyushu	6
Ps	[Position]	Dean or similar, Prof., Associate prof., Lecturer, Assistant prof.	5
Gn	[Generation] year of birth	T1 (–1934), T2 (1935–1946), T3 (1947–1949), T4 (1950–1964), T5 (1965–1969), T6 (1970–1986)	6
Dg	[Degree]	Doctor, Master, Bachelor, Other	4
Gu	[Graduating univ.]	Own univ., Overseas univ., National univ. (G1, G2, G3, G4), Public univ., Private univ., National academy/other	9
Rigu	[Region in graduating univ.]	Kanto, Chubu, Kinki, Overseas, Chugoku/Shikoku, Hokkaido/Tohoku, Kyushu	7
Fl	[Field] academic fields	Multidisciplinary, New Composite Disciplines, Humanities, Social Science, Mathematics, Physical Sciences, Biology, Chemistry, Engineering, Agriculture, Medicine/Dentistry/Pharmacology	11
#t	[# of transfers] number of moves between univs.	0, 1, 2, 3, 4+	5
#p	[# of promotions] number of promotions during observation period	0, 1, 2, 3, 4+	5
Pp	[Promotion path] combination of positions from beginning to end of observation period	e.g. from Associate prof. to Prof.	20

Figure 2 in the Appendix 1 gives the frequencies for all fields. As can be seen in models M1 to M4 in Table 3 in Appendix 1, ZIP and ZINB are also superior to PO and NB.

Characteristics in the optimal all-fields model

Next, let us examine the effects of the professors' covariates on getting the grants. For this purpose, we consider the generalized linear models (GLM) of PO and NB shown in Table 1. See Agresti (1996, 2002), Annette (2002), Brian and Torsten (2010), and Joseph (2011) for a detailed description of GLM.

For the GLMs of PO, NB and ZIP, the mean values were regressed with dummy variables based on the applicants' covariates. Our preliminary study indicated that a model including all the covariates is unstable and lacks statistical significance. Therefore, we used the Bayesian information criterion (BIC) for deriving a smaller but significant subset of

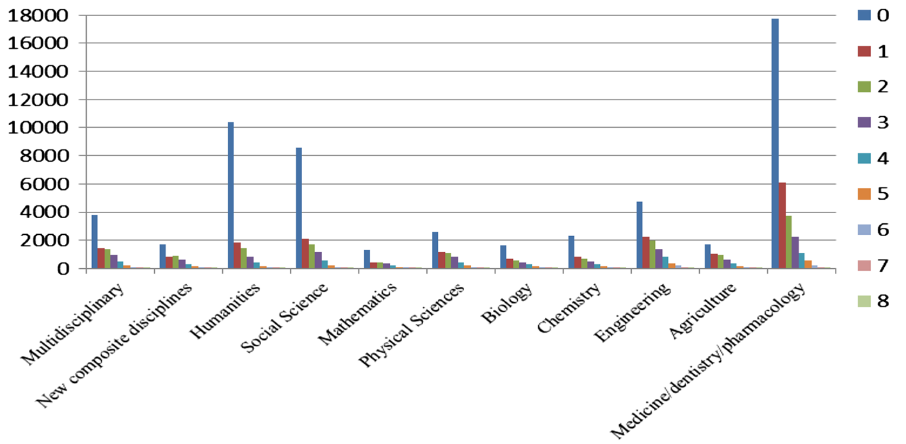

Fig. 1 Frequencies by field of the number of awarded researchers for the eight observation time points

Table 2 Optimal model among PO, NB, ZIP and ZINB and its zero-inflate probability by field

Field Sample size	Multidisciplinary $n = 8490$	New Composite Dis. $n = 4628$	Humanities $n = 15{,}165$	Social Sci. $n = 14{,}567$	Mathematics $n = 2953$	Physical Sci. $n = 6557$
Opt. model	ZINB	ZIP	ZINB	ZINB	ZIP	ZINB
Zero-inflated probability	0.374	0.282	**0.618**	**0.520**	0.384	0.323

Field Sample size	Biology $n = 3890$	Chemistry $n = 4986$	Engineering $n = 11{,}879$	Agriculture $n = 5087$	Medicine/Dentistry/Pharmacol. $n = 31{,}854$
Opt. model	ZINB	ZINB	ZINB	ZINB	ZINB
Zero-inflated probability	0.349	0.406	0.322	0.243	**0.465**

covariates. The BIC was evaluated by making an exhaustive search of $2^{13} = 8192$ formulas with main-effect models. The PO, NB and ZIP regressions are listed in Table 3. We also examined multinomial logistic (MNL) regression for all fields. See for "Selection of multinomial logistic regression models" for the model. The best model for all fields was M13 whose BIC is denoted in bold face. The selected two-factor interactions of the optimal models are listed in Appendix 2.

The best model (M12 of Table 3) for predicting the number of awards by using the characteristics of researchers in all fields is summarized as follows.

Fy and #op — No effect. This means that the grants are stable for the observation duration from 1988 to 2006

Af	G1 > G2 > G3 > Private univ. > G4 = Public univ. > National academy/other The above line means that the effect of G1 universities is largest, that of G2 universities comes next, and so on. The equality "G4 = Public univ." means that the award effects of G4 universities and Public universities are the same
Riu	No effect. The location of the universities is independent of the award.
Ps	No effect. The position effect is not clear for the all-fields data. However, it is clear for most of the individual fields
Gn	T6 > T5 > T4 > T3 = T2 > T1 The youngest generation is T6, the second youngest generation is T5, and so on. The order of the generation effects means that younger generations have a greater chance to be awarded. One of the reasons is a preferential treatment system for young researchers started in 2002
Dg	Doctor > Master = Other > Bachelor. The research ability may be proportional to the level of degrees
Gu and Rigu	No effect. The grant application form does not have a column for indicating graduating universities
Fl	Medicine/Dentistry/Pharmacology > Agriculture > Biology = Chemistry = Engineering > Multidisciplinary = Physical Sciences > New Composite Disciplines = Mathematics > Humanities = Social Science. The reason why the medical fields have the largest effect was described in Section "Features of eleven academic fields in terms of zero-inflation probability"
#t	1+ > 0. Researchers who had made transfers are more likely to be awarded. However, transfer is a result, not a cause of getting the grants
#p and Pp	No effect. The number of promotions and promotion paths are independent of awarding. However, getting grants is one of the considerations that go into making decisions as to which professors move on to higher academic positions

The coefficient values estimated for the above are listed in Table 4 of Appendix 1.

Optimal model by field

This section derives the optimal models for individual academic fields. The results reveal interesting features of the individual fields despite that the decisions regarding grants are supposed to be independent of the academic field.

Multidisciplinary field with the optimal ZIP model with 20 df (n = 8490)

The optimal model is the ZIP model with df 20, which has the following effects; others were not detected.

Af	G1 > G2 > G3 = Private univ. = National academy/other > G4 = Public univ.
Ps	Dean or similar > Prof. > Associate prof. = Lecturer = Assistant prof.

Gn T6 > T5 > T4 = T3 > T2 = T1

The optimal model with df 20 is the simplest among the 11 fields. In this case, no two-factor interaction was detected. Multidisciplinary fields unify various academic disciplines. Therefore, this is why the effect of the affiliation G1–G2 (active universities having many faculties) is large for this field.

New Composite Disciplines with the optimal ZIP model with 23 df (n = 4628)

The optimal model has the following effects.

Af G1 > G2 > G3 > G4 = Public univ. = Private univ. = National academy/other
Ps Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof.
Gn T6 > T5 > T4 = T1 > T3 = T2
Dg Doctor > Master = Other > Bachelor
#t 3 > 1 = 2 > 0 > 4+

The optimal ZIP model is the second simplest with df 23, which is the feature of this field as well as the Multidisciplinary field. No two-factor interaction was detected. The applicants of this field must have various academic disciplines. Therefore, the reason why the effect of "# of transfers" is large may have come from the likelihood of the awarded applicants previously belonging to several institutions.

Humanities with the optimal ZIP model with 44 df (n = 15,165)

The optimal model has the following effects.

Af G1 > G2 = Private univ. > G3 = G4 = National academy/other > Public univ.
Ps Dean or similar > Prof. > Associate prof. = Assistant prof. > Lecturer
Gn T6 > T5 > T4 > T3 = T2 = T1
Dg Doctor > Master > Bachelor = Other
#t 2 = 3 > 1 > 0 = 4+

Japanese universities hire many lecturers for teaching foreign languages. The fact that the Lecturer's main effect is the smallest among the position effects may reflect this special character of Humanities. Also, the main effect of two or three transfers is highest because fluidity of applicants is high. Many faculty members in Humanities now have doctoral degrees. This is one reason why two-factor interactions were detected (Table 5). From the interactions, the effect of a doctoral degree is the largest among the degree effects. However, the magnitude of the effect depends on the generation or affiliation.

Social Science with the optimal MNLogit model with 136 df (n = 14,567)

The optimal MNLogit model has the following effects.

Af G1 > G2 > G3 = G4 = Private univ. > Public univ. = National academy/other
Gn T6 > T5 > T1 > T4 > T3 = T2
Dg Doctor > Master > Bachelor = Other

The MNLogit model without two-factor interaction is optimal. This implies that "Social Science" has a time-dependent structure.

Mathematics with the optimal ZIP model with 27 df (n = 2953)

The optimal model has the following effects.

Af	G1 = G2 = Private univ. > G3 = G4 = Public univ. > National academy/other
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof.
Gn	T6 > T5 > T4 > T3 = T2 > T1
Dg	Doctor > Bachelor = Master > Other

The optimal model is simple, and no two-factor interaction was detected. The effects of G1, G2, and private univ. are the same. This may be due to there being especially equal opportunity for researchers to be awarded in Mathematics.

Physical Sciences with the optimal ZIP model with 29 df (n = 6557)

The optimal model has the following effects.

Af	G1 > G2 = Private univ. = G3 = G4 = Public univ. > National academy/other
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof.
Gn	T6 > T5 > T4 = T3 = T2 = T1
Dg	Doctor > Master = Other > Bachelor

This is similar to the case of Mathematics. From the interactions (Table 6), the award effect of generations T1–T4 with doctoral degrees is large, while the effect is small in T5–T6 generations. This is because most young applicants in this field have doctoral degrees.

Biology with the optimal ZIP model with 34 df (n = 3890)

The optimal model has the following effects.

Af	G1 = G2 > Public univ. = Private univ. = National academy/other > G3 = G4
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof
Gn	T6 > T5 = T1 > T4 = T3 = T2
Dg	Doctor > Master = Other > Bachelor
Rigu	Kanto > Overseas > Chubu = Kinki > Hokkaido/ Tohoku = Kyushu > Chugoku/Shikoku
#t	1+ > 0

The main effects of G1 and G2 are the same, which may come from severe competition in this field. Also, the overseas' effect of region in graduating university comes second largest in the main effect of Rigu, which suggests the field is activated by foreign countries. The interactions (Table 7) indicate that applicants who have transferred at least once are more likely to be awarded even if they do not belong to G1 or G2 and have only Bachelor degrees.

Chemistry with the optimal ZIP model with 40 df (n = 4986)

The optimal model has the following effects.

Af	G1 > G2 = Private univ. > G3 = G4 = Public univ. = National academy/other
Gn	T6 > T5 > T4 = T1 > T3 = T2
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof.
Dg	Doctor > Master = Other > Bachelor

Rigu	Kanto = Kinki = Overseas > Hokkaido = Tohoku = Chubu = Kyushu > Chugoku/Shikoku
#t	1+ > 0
#p	0 > 1 > 2+

The main effect of private universities is equivalent to that of G2. The reason seems to be that chemistry, even experimental chemistry, is an inexpensive field of study. From the interactions (Table 8), applicants having transferred at least once are likely to be awarded if they were affiliated with G3, G4, Public universities, or National academy/other. This may indicate that the awarded applicants had moved from G1 or G2 universities.

Engineering with the optimal ZIP model with 48 df (n = 11,879)

The optimal model has the following effects.

Af	G1 > G2 > Private univ. > G3 = G4 = Public univ. > National academy/other
Gn	T6 > T5 > T4 > T3 = T2 = T1
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof.
Dg	Doctor > Master > Bachelor = Other
#t	0 = 3+ > 1 = 2
#p	0 > 1 > 2+

Engineering covers wide academic and practical ranges. Therefore, it required the biggest model among the ZIP models (Table 9). Private universities play a large role as do G2 universities because most of the private science universities have a faculty of engineering, even if they do not have a faculty of science.

Agriculture with the optimal ZIP model with 28 df (n = 5087)

The optimal model has the following effects.

Af	G1 > G2 = Private univ. > G3 = Public univ. > G4 > National academy/other
Ps	Dean or similar > Prof. > Associate prof. > Lecturer = Assistant prof
Gn	T6 > T5 > T4 = T3 > T2 = T1
Dg	Doctor > Master = Other > Bachelor

The main effect indicates that private universities play as large a role as G2 universities do. The number of transfers did not have a relationship with awarding. This comes from the fact that the fluidity of Agriculture is the lowest because it is not easy for researchers to move his/her experimental field to another institution. From the interactions (Table 10), the award effects of Dean or similarly of generations T1–T2 and Associate prof. of T6 are large. This two-pole tendency is unique among the fields.

Medicine/Dentistry/Pharmacology with the optimal MNLogit model with 144 df (n = 31,854)

The optimal model, MNLogit model, has the following effects.

Af	G1 > G2 > G3 > G4 = Public univ. = Private univ. = National academy/other
Ps	Dean or similar > Prof. > Associate prof. = Lecturer = Assistant prof.
Gn	T6 > T5 > T4 > T3 = T2 = T1
Dg	Doctor > Bachelor = Other > Master

No two-factor interaction was detected. It can be seen here that G4, public, and private universities and academies are integrated, which suggests a relationship with regional university hospitals, and that junior faculty members can be treated as one group. This is because the power of professors in these fields might be much larger than in other fields. Also, the effect of "Master degree" is the lowest of any field. The reason is that Japanese universities and national academies tend have large hospitals and play active roles in regional general medicine.

Concluding remarks

This study is based on publically available information on representative researchers and the grants they have received. It investigated the relation between the number of grants awarded and the applicants' attributes. Of course, research proposals are screened in a peer-review process, and roughly speaking, 25 % of them are awarded funds. In examining the numbers of applicants and awarded applicants by field, we observed that there are many zero-count (zero-inflate) cases. We used 13 attributes as covariates for generalized linear models. The following are our findings.

1. We detected zero-inflation in all the fields as well as individual fields, and the magnitudes of zero-inflation give an important feature of the fields. Those zero-inflate cases may come from the situation that the primary duty of the applicants is teaching—the number of such faculty members is substantial. Investigating this phenomenon is meaningful from an academic and scientific point of view, but it is hardly meaningful in terms of setting national policy.
2. When we examined the all fields and each field individually, we found that applicants belonging to the most active universities and having doctoral degrees are likely to get funds. This is also true for young researchers and applicants holding higher-level positions. It is also quite interesting that applicants who have transferred at least once are also likely to be awarded. Most of the main effects are similar by field, but the selected two-factor interactions show the characteristics of the eleven fields.

The main purpose of this study was to clarify how grants affect researchers and are affected by researchers and to contribute to the formulation of useful policies. It was quite surprising and interesting that features of academic disciplines were detected through the statistical models despite that policy makers are supposed to give equal treatment to applications from any discipline.

In future studies, we plan to make improvements to our models by carefully selecting the covariates and/or levels. Japanese policy makers involved in the grant system should know the features of each discipline including zero-inflation. Furthermore, the disciplines should be respected and treated as independent from one another.

Acknowledgments The authors are very grateful for the reviewers giving constructive comments which led to a much better revision. We are also grateful to Kojunsha, the publishers of the directory listing the staff of Japanese universities. This research was partly supported by JSPS KAKENHI Grant Numbers JP23300106 and JP15H02670.

Appendix 1: The optimal model for all fields

Comparison of distributions of number of awards in all fields and in each field

First, let us examine the distribution of the numbers of awards in all fields during 1988–2006. The raw data in Fig. 2 illustrates that 56,572 professors (Prof.) failed to get a

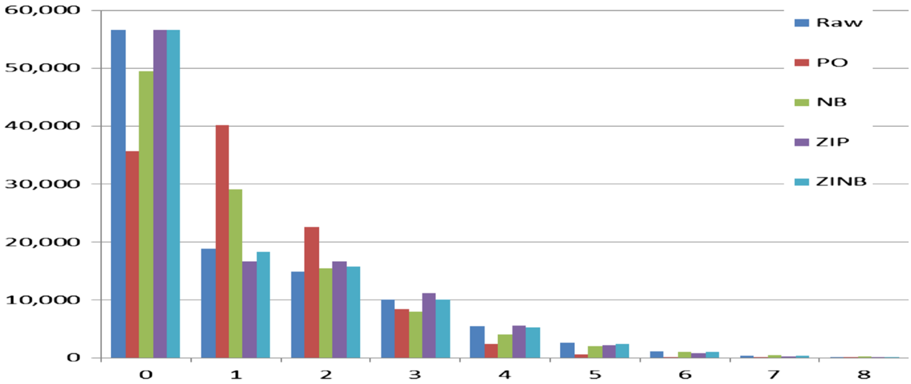

Fig. 2 Frequencies of number of awarded professors and predicted values for all fields

Table 3 The optimal models with/without level integration and category selection in all fields

Model No.	Distribution	Model selection	Category of covariates	Zero-inflate probability	df	BIC
M1	PO	–	Not used	Not used	1	358,089
M2	NB	–	Not used	Not used	2	323,129
M3	ZIP	–	Not used	0.4384	2	318,749
M4	ZINB	–	Not used	0.4058	3	318,080
M5	PO	Exhaustive	–Ps	Not used	84	276,504
M6	NB	Exhaustive	–Riu –Ps	Not used	79	273,652
M7	ZIP	Exhaustive	–Rigu –Ps	0.1678	65	271,823
M8	MNLogit	Forward	–Ps –Gu	Not used	648	269,116
M9	MNLogit	Exhaustive	–Riu –Gu –Rigu –#p –Pp	Not used	376	264,530
M10	ZIP	Level Integration of M5		0.1741	37	272,701
M11	ZINB	Level Integration of M6		0.1741	38	272,712
M12	MNLogit	Level Integration of M9		Not used	289	**263,693**

Note that "–Ps" in model M5 denotes that the covariate [Position] is excluded from the optimal model

grant as representative researchers, 18,843 professors were awarded once among eight of the observation time points. We fitted a Poisson (PO) distribution to the data, and the red bars in the figure are the resulting theoretical frequencies. It can be seen that the PO distribution is not appropriate for predicting the zero frequency. Although a negative binomial (NB) distribution improves the fit (the green bars in Fig. 2), the estimate for one time is poor. We therefore fitted zero-inflated distributions, defined as follows.

Let Y_0 be a PO random variable. Then, a random variable Y follows a zero-inflated PO distribution (ZIP) if its distribution is given by $\mathrm{P}(Y = 0) = \nu + (1 - \nu)\mathrm{P}(Y_0 = 0)$ and $\mathrm{P}(Y = k) = (1 - \nu)\mathrm{P}(Y_0 = k)$ for $k = 1, 2, \ldots$.

The parameter $0 < v < 1$ is called a zero-inflate probability. A zero-inflated NB distribution (ZINB) is similarly defined.

PO, NB, ZIP and ZINB were fitted to the data and evaluated using BIC, as it derives a significant but simpler distribution than AIC does. A comparison of the first four models in Table 3 as to the effect of zero-inflation, where df means the degree of freedom of the models, led us to conclude that the zero-inflated models, ZIP and ZINB, are superior to PO and NB. Also, the respective zero-inflate probabilities are large.

The distribution by research field in the all-fields model is shown in Table 3. Unfortunately, the parameter estimation of the ZINB regression failed to converge.

The three BIC values of models M6–M8 are much smaller than those of M1–M4 (Table 2). This means that getting grants is highly dependent on the covariates of the applicants. Among the first seven models, ZIP regression is optimal.

Selection of multinomial logistic regression models

A more flexible model including zero-inflation can be considered. It is a multinomial logistic regression model (MNLogit) in which the number of awards ($Y = 0, 1, \ldots, 8$) is taken to be a group label. Suppose that the i-th faculty member has p attributes (covariates) $\boldsymbol{x}_i = (x_{i1}, \ldots, x_{ip})^t$, where x_{ij} take zero or one according to whether the member has an attribute (level) j or not. Accordingly, the MNLogit model is based on a multinomial distribution with the following regression:

$$\log\{P(Y = k|x_i)/P(Y = 0|x_i)\} = \beta_{k0} + \beta_{k1}x_{i1} + \cdots + \beta_{kp}x_{ip} \text{ for } k = 1, 2, \ldots, 8.$$

MNLogit regressions based on forward selection and exhaustive selection of the covariates are evaluated using BIC in the 8-th and 9-th rows of Table 2. An exhaustive search found a better model with a small BIC value. The target variable [# of grants awarded] can be regressed using the covariates as follows:

$$\begin{aligned}&[\#\text{of grants awarded}] \sim [\text{Final year}] + [\#\text{of observation points}] + [\text{Affiliation}]\\&\quad + [\text{Generation}] + [\text{Position}] + [\text{Degree}] + [\text{Field}] + [\#\text{of transfers}]\end{aligned}$$

Level integration and selection of categories of covariates

Next, let us integrate the levels of covariates whose estimated coefficients are nearly equal. In order to avoid a structural problem, we exclude the two categories of [Final year] and [# of observation points]. The level integration process consists of the following steps:

Level integration process

1. Sort levels within the same category according to the estimated coefficients for the number of grants awarded.
2. Integrate the levels between which the difference in effect (difference in estimated value) is small.
3. Repeat steps (1) and (2) until BIC begins to increase.

After the level integration, BIC is used to select the remaining eleven categories.

Level integration of ZIP and ZINB regressions

We can study the idea of whether integrating levels in the ZIP optimal model can do better than this MNLogit model.

[Affiliation]	G1 > G2 > G3 = Private univ. > G4 = Public univ. > National academy/other
[Region in univ.]	Kanto = Chubu > Kinki > Hokkaido = Tohoku = Chugoku/Shikoku = Kyushu
[Generation]	T6 > T5 > T4 > T3 = T2 = T1
[Graduating univ.]	G1 > Own univ. > G2 = G3 > Public univ. = foreign univ. > G4 = Private univ. > National academy/other
[Field]	Medicine/Dentistry/Pharmacology > Biology = Agriculture > Chemistry = Engineering > Multidisciplinary = New Composite Disciplines = Physical Sciences > Mathematics > Social Science > Humanities
[# of transfers]	2 = 3 > 1 > 4+ > 0
[# of promotions]	0 > 1 > 2 = 3 > 4+

Note that G3 = Private univ. in [Affiliation] means that two levels, G3 and Private universities, were integrated. This implies that G3 and Private universities provided the same effect for awarding. The order of five integrated levels in [Affiliation] means that G1 universities had the largest positive effect for awarding, G2 universities come next, and so on.

Level integration to MNLogit models

Regarding the MNLogit model (M9 of Table 3), five categories were selected and level integration was examined. This integration causes the BIC to drop to 263,693 (M12 of Table 3), and hence, M12 is the best model in the all-fields case. The estimated coefficients are listed in Table 4.

The optimal multinomial logistic model for all fields after the level integration is given by:

[Affiliation]	G1 > G2 > G3 > Private univ. > G4 = Public univ. > National academy/other
[Generation]	T6 > T5 > T4 > T3 = T2 > T1
[Degree]	Doctor > Master = Other > Bachelor
[Field]	Medicine/Dentistry/Pharmacology > Agriculture > Biology = Chemistry = Engineering, Multidisciplinary = Physical Sciences > New Composite Disciplines = Mathematics > Humanities = Social Science
[# of transfers]	1+ > 0

Table 4 Estimated coefficients of the optimal logistic model for all fields

Freq.	Intercept	Final year							
		1988	1991	1994	1997	2000	2003	2005	2006
1	0.34	0.00	−0.59	−0.46	−0.48	−0.40	−0.44	−0.22	−0.50
2	−1.11	0.00	−0.06	−0.47	0.08	−0.02	−0.01	0.36	0.77
3	−1.26	0.00	−3.58	0.15	0.31	0.26	0.28	0.69	1.17
4	−0.80	0.00	0.20	0.14	−0.28	−0.51	−0.60	−0.39	0.37
5	−1.23	0.00	0.24	−0.19	0.11	0.46	−0.47	0.25	0.96
6	−0.85	0.00	0.21	−0.29	−0.24	−0.38	0.22	0.09	0.19
7	−1.54	0.00	−0.02	−0.37	−0.74	−0.59	−0.08	0.75	0.38
8	−4.26	0.00	0.05	−0.61	−1.40	−0.73	−0.45	−0.09	0.30

Freq.	Number of observation points							
	1	2	3	4	5	6	7	8
1	−0.89	−0.20	−0.25	−0.10	−0.08	−0.08	−0.03	0.00
2	−3.03	−0.69	−0.12	−0.19	−0.19	0.10	0.14	0.00
3	2.37	−11.25	−0.65	−0.38	−0.06	−0.12	0.11	0.00
4	0.13	−0.26	−4.25	−0.77	−0.43	−0.09	0.15	0.00
5	0.18	0.55	2.69	−6.99	−0.70	−0.36	−0.05	0.00
6	0.18	−0.20	0.05	−0.01	−2.64	−1.23	−0.31	0.00
7	−0.01	−0.45	−0.33	0.02	0.23	−2.96	−0.66	0.00
8	0.24	0.13	−0.39	0.09	0.47	0.04	−1.98	0.00

Freq.	Affiliation						Generation				
	G1	G2	G3	G4 = Public univ.	Private univ.	National academy/other.	T1	T2 = T3	T4	T5	T6
1	0.00	−0.29	−0.65	−0.81	−0.53	−1.72	−0.23	−0.53	0.00	0.48	0.75
2	0.00	−0.39	−0.98	−1.15	−1.05	−1.89	−0.15	−0.47	0.00	0.64	1.28

Table 4 continued

Freq.	Affiliation						Generation				
	G1	G2	G3	G4 = Public univ.	Private univ.	National academy/other.	T1	T2 = T3	T4	T5	T6
3	0.00	−0.41	−1.22	−1.44	−1.31	−2.09	−0.16	−0.59	0.00	0.89	1.63
4	0.00	−0.51	−1.36	−1.72	−1.42	−2.30	−0.38	−0.53	0.00	1.02	1.74
5	0.00	−0.75	−1.56	−2.04	−1.58	−2.43	−3.22	−0.39	0.00	0.89	1.38
6	0.00	−0.59	−1.50	−2.07	−1.78	−2.25	−1.43	−0.20	0.00	1.14	1.66
7	0.00	−0.67	−1.67	−2.08	−1.95	−1.98	−1.38	−0.03	0.00	1.55	2.01
8	0.00	−1.19	−1.58	−2.18	−2.56	−6.43	−0.47	0.17	0.00	1.01	1.21

Freq.	Position					Number of transfers	
	Assistant prof.	Lecturer	Associate prof.	Prof.	Dean or similar	0	1+
1	−0.61	−0.55	−0.32	0.00	0.64	0.00	0.14
2	−1.26	−1.06	−0.60	0.00	0.56	0.00	0.32
3	−1.66	−1.46	−0.78	0.00	1.17	0.00	0.50
4	−2.03	−1.96	−1.11	0.00	1.07	0.00	0.54
5	−2.44	−2.14	−1.51	0.00	1.02	0.00	0.54
6	−3.05	−3.64	−1.69	0.00	1.03	0.00	0.57
7	−2.79	−3.19	−1.84	0.00	1.23	0.00	0.42
8	−2.32	−1.40	−1.92	0.00	1.53	0.00	1.25

Freq.	Degree			Field					
	Bachelor	Master = Other	Doctor	Engineering Chemistry Biology	Humanities Social Science	Mathematics New Composite Disciplines	Multidisciplinary = Physical Sciences	Agriculture	Medicine/ Dentistry/ Pharmacology
1	0.00	0.20	0.99	−0.31	−0.82	−0.37	−0.44	−0.09	0.00
2	0.00	0.65	1.65	−0.33	−0.92	−0.48	−0.50	0.02	0.00

Table 4 continued

Freq.	Degree			Field					
	Bachelor	Master = Other	Doctor	Engineering Chemistry Biology	Humanities Social Science	Mathematics New Composite Disciplines	Multidisciplinary = Physical Sciences	Agriculture	Medicine/ Dentistry/ Pharmacology
3	0.00	0.42	1.55	−0.38	−1.17	−0.59	−0.61	0.00	0.00
4	0.00	0.55	1.89	−0.40	−1.34	−0.69	−0.77	−0.06	0.00
5	0.00	0.09	1.64	−0.43	−1.78	−0.83	−0.76	−0.14	0.00
6	0.00	−0.32	1.45	−0.47	−1.95	−1.23	−1.03	0.06	0.00
7	0.00	−0.27	1.34	−0.71	−2.31	−2.10	−1.50	−0.69	0.00
8	0.00	0.94	1.82	−0.54	−2.07	−1.84	−0.64	−0.53	0.00

Appendix 2: The optimal models up to two-factor interactions by field

The estimated values of the two-factor interactions for humanity, physical sciences, biology, chemistry, engineering and culture are listed in Table 5–10).

Table 5 Two-factor interactions of ZIP for humanities

Generation	Affiliation		
	G1 = G2 = Private univ.	G3 = G4 = National academy/other	Public univ.
T1 = T2 = T3	*0.000*	−0.362	−0.613
T4	*0.000*	*0.000*	*0.000*
T5	*0.000*	0.175	0.215
T6	*0.000*	0.176	0.192
Generation	**Degree**		
	Bachelor = Other	Master	Doctor
T1 = T2 = T3	*0.000*	0.188	0.581
T4	*0.000*	*0.000*	*0.000*
T5	*0.000*	0.974	0.912
T6	*0.000*	0.215	0.154
Degree	**Affiliation**		
	G1 = G2 = Private univ.	G3 = G4 = National academy/other	Public univ.
Bachelor = Other	*0.000*	*0.000*	*0.000*
Master	*0.000*	0.360	−0.376
Doctor	*0.000*	0.493	0.136

Table 6 Two-factor interactions of ZIP for physical sciences

Degree	Generation		
	T1 = T2 = T3 = T4	T5	T6
Bachelor	*0.000*	*0.000*	*0.000*
Master = Other	1.616	*0.000*	−0.264
Doctor	2.422	*0.000*	–

Table 7 Two-factor interactions of ZIP for biology

Number of transfers	Affiliation		
	G1 = G2	Public univ. = Private univ. = National academy/other	G3 = G4
1+	*0.000*	0.241	0.281
0	*0.000*	*0.000*	*0.000*

Number of transfers	Degree		
	Bachelor	Master = Other	Doctor
1+	*0.000*	−0.873	−1.761
0	*0.000*	*0.000*	*0.000*

Table 8 Two-factor interactions of ZIP for chemistry

Number of transfers	Affiliation		
	G1	G2 = Private univ.	G3 = G4 = Public univ. = National academy/other
1+	*0.000*	0.090	0.476
0	*0.000*	*0.000*	*0.000*

Generation	Position			
	Lecturer = Assistant prof.	Associate prof.	Prof.	Dean or similar
T1 = T4	−0.650	−0.343	*0.000*	–
T2 = T3	−1.251	−1.050	*0.000*	−0.038
T5	*0.000*	*0.000*	*0.000*	*0.000*
T6	0.393	0.227	*0.000*	–

Table 9 Two-factor interactions of ZIP for engineering

Number of transfers	Affiliation				
	G1	G2	G3 = G4 = Public univ.	Private univ.	National academy/other
0 = 3+	*0.000*	0.058	0.225	−0.005	1.334
1 = 2	*0.000*	*0.000*	*0.000*	*0.000*	*0.000*

Generation	Position			
	Lecturer = Assistant prof.	Associate prof.	Prof.	Dean or similar
T1 = T2 = T3	−1.094	−0.841	*0.000*	0.019
T4	*0.000*	*0.000*	*0.000*	*0.000*
T5	0.525	0.205	*0.000*	–

Table 9 continued

Generation	Position			
	Lecturer = Assistant prof.	Associate prof.	Prof.	Dean or similar
T6	0.858	0.414	*0.000*	–

Degree	Position			
	Lecturer = Assistant prof.	Associate prof.	Prof.	Dean or similar
Bachelor = Other	*0.000*	*0.000*	*0.000*	*0.000*
Master	−0.396	−0.242	*0.000*	–
Doctor	−1.309	−1.148	*0.000*	–

Number of transfers	Number of promotions		
	0	1	2+
0 = 3+	*0.000*	−0.205	−0.189
1 = 2	*0.000*	*0.000*	*0.000*

Table 10 Two-factor interactions of ZIP for agriculture

Generation	Position			
	Lecturer = Assistant prof.	Associate prof.	Prof.	Dean or similar
T1 = T2	−1.183	−1.089	*0.000*	0.038
T3 = T4	−0.302	−0.296	*0.000*	–
T5	*0.000*	*0.000*	*0.000*	*0.000*
T6	−0.031	0.011	*0.000*	–

References

Agresti, A. (1996). *An introduction to categorical data analysis*. NJ: Wiley.

Agresti, A. (2002). *Categorical data analysis* (2nd ed.). NJ: Wiley.

Annette, J. D. (2002). *An introduction to generalized linear models* (2nd ed.). UK: Chapman & Hall.

Brian, S. E., & Torsten, H. (2010). *A handbook statistical analyses using R* (2nd ed.). London: Taylor and Francis Group LLC.

Fukuzawa, N. (2014). An empirical analysis of the relationship between individual characteristics and research productivity. *Scientometrics, 99*, 785–809.

Hosotsubo, M. (2011). A statistical study of transferal and promotion mechanisms relating to the appointment of professors at Japanese national universities based on cross tabulation and log-linear model analysis. *Scientometrics, 86*(2), 405–430.

Ida, T., & Fukuzawa, N. (2013). Effects of large scale research funding programs a Japanese case study. *Scientometrics, 94*, 1253–1273.

Joseph, M. H. (2011). *Negative binomial regression* (2nd ed.). Cambridge: Cambridge University Press.

Rao, I. K. R. (1980). The distribution of scientific productivity and social change. *Journal of the American Society for Information Science, 31*, 121.

Ripley, B., & Venables, W. (2015). *R: nnet package*. http://cran.r-project.org/web/packages/nnet/nnet.pdf

Stasinopoulos, M., Rigby, B., Voudouris, V., Akantziliotou, C., & Enea, M. (2015). *R: gamlss package*. http://cran.r-project.org/web/packages/gamlss/gamlss.pdf

Yamasaki, H. (1982). Scientific productivity and its division-by-class process—Basic theory of the "Matthew effect". Daigaku Ronshu Research in Higher Education (Vol. 11, pp. 1–21). Research Institute for Higher Education, Hiroshima University. (**in Japanese**)